

Yu-Zheng Lin, Ph.D.

Lawful Permanent Resident in the U.S.

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PROFESSIONAL SUMMARY

Ph.D. in Electrical and Computer Engineering from the University of Arizona, with a strong research record in digital twins, cyber-physical systems, large language models, and AI-driven intelligent systems. First author of an ACL 2026 Main Conference paper, ranked in the top 50% of accepted papers, with publications in IEEE TAI, ACM TODAES, and Springer Cybersecurity. Research focuses on probabilistic modeling, dynamic data-driven systems, and LLM-based reasoning under uncertainty, with applications in digital twins, cybersecurity, and human-centered AI.

EDUCATION

2022 - Present	The University of Arizona , Tucson, AZ Ph.D. Candidate – Electrical and Computer Engineering (Anticipated graduation date: May 15) – Research Interest: High-Performance Computing, Agentic AI, Digital Twin, Signal Processing, Machine Learning, Large Language Model, Cybersecurity – Advisor: Dr. Pratik Satam
2016	National Taiwan University of Science and Technology (TaiwanTech) , Taiwan M.Sc. – Electronics and Computer Engineering
2014	Hsiuping University of Science and Technology , Taiwan B.E. – Electrical Engineering

PROFESSIONAL EXPERIENCE

2022 Aug - Present	Graduate Research Assistant (NSF Funded), The University of Arizona
2016 Nov – 2019 Dec	Research Assistant , Instrumentation Development Division – Vacuum Group, National Synchrotron Radiation Research Center, Taiwan
2014 Sep - 2016 Aug	Graduate Research Assistant , Photonics System and Simulation Design Laboratory, National Taiwan University of Science and Technology, Taiwan
2013 July - 2013 Sep	Internship, Department of Research and Development SYRIS Technology Corp., Taiwan
2012 Aug - 2014 Aug	Undergraduate Research Assistant , Wireless Sensing Laboratory, HUST, Taiwan

ADDITIONAL TRAINING

2025 June	CI Pathway – ML& AI Track (Competitively selected participant) The National Center for Supercomputing Applications, UIUC , Certification Link - Focused on machine learning workflows on HPC systems - Conducted experiments on NCSA’s Delta supercomputer to optimize large-scale ML tasks - Hands-on training with PyTorch, model parallelism, and multi-GPU training techniques
2024 July	CoDaS-HEP: Computational and Data Science for High Energy Physics , Princeton University , with Full Travel Funding, Link Focus on high-performance computing topics, including machine learning, parallel computing, and data processing.
2024 April	AI for Science on Supercomputer Argonne National Laboratory , Certification Link - Gained foundational understanding of the mathematical principles underpinning machine learning algorithms - Gained hands-on experience submitting and executing parallel jobs on Polaris, a 44-petaflop supercomputer at the Argonne Leadership Computing Facility

RESEARCH FUNDING

2025	NSF ACCESS Explore (Principal Investigator) - 200,000 credits - Project name: Toward Resilience in Cyber-Physical Systems: A Generative Digital Twin–Enabled Dynamic Data-Driven Framework for Decision-Making under Uncertainty - Equivalent to \$5,000 based on CloudBank Exchange Rates
2025	Open AI Researcher Access Program (Principal Investigator) - \$2000 For large language model robustness studies and exploring potential applications in knowledge discovery and data mining.

AWARDS

Highlight

2018 **Isamu Abe Prize**, The Isamu Abe Prize recognizes innovative ideas, achievements, and applications in the field of accelerator control, and it is granted every two years by PCaPAC.

Others

2024 **Outstanding Teaching Assistant**, The University of Arizona

2017 **The Silver Medal Award**, Taipei International Invention Show & Technomart

2015 **3rd Place**, The 6th NTUST Entrepreneurship Competition (TaiwanTech) - Optics Brick

2014 **1st Place**, The 9th Digital Signal Processing Creative Design Contest Award - Healthcare Applications (Ministry of Education (Taiwan))

2013 **2nd Place & Texas Instruments Award**, The 8th Digital Signal Processing Creative Design Contest Award - Green Energy and Control (Ministry of Education (Taiwan))

2012 **Bronze Medal Award**, TEMI International Microcontroller Technical and Creative Application Contest - Mini Self-Driving Car Racing (Taiwan Embedded Microcontroller Development Institute)

2012 **An Excellent Work**, The 2th NDHU Solar Energy & Photonics Creativity Contest Industry-University Cooperative (National Dong Hwa University - Energy Technology Center)

2012 **Remarkable Award**, The 7th Holtek MCU Creativity Contest - Control Application (Holtek Semiconductor)

RESEARCH INTERESTS & EXPERTISE

Research Interests

- Machine Learning, Large Language Model, Signal Processing, Digital Twin, Agentic AI, Human-centered Systems
- Cyber Security - Cyber-Physical System Security, ML-based intrusion detection
- Internet of Things

Techniques

- Programming Skills: C/C++, Python, JavaScript (Front-End & Back-End)
- Machine Learning: Tensorflow, PyTorch, Sci-Kit Learn, Large Language Models, RAG (GraphRAG, LangChain)
- High Performance Computing: OpenMP
- Data Engineering & Visualization: Databases (InfluxDB, MongoDB, MySQL), Grafana
- Cloud Platform (GCP, AWS)
- Linux, Git

TEACHING EXPERIENCES

2024 Fall **Teaching Assistant**, Web Development and the Internet of Things

2021 Summer **Professional Guest Lecturer from Industry**,
MLOps Engineering: Building Reliable ML Systems

2014 Fall **Teaching Assistant**, Introduction to Photonics Engineering Laboratory

2014 Spring **Teaching Assistant**, Introduction to Digital Signal Processing

2013 Fall **Teaching Assistant**, Principles of Microprocessor Systems

PROFESSIONAL REFERENCES

Prof. Pratik Satam	Prof. Soheil Salehi	Prof. Sicong Shao
University of Arizona	University of Arizona	University of Arizona
• Relationship: Ph.D Advisor	• Relationship: Major Collaborator	• Relationship: Major Collaborator
• Email: pratiksatham@arizona.edu	• Email: ssalehi@arizona.edu	• Email: sicong.shao@UND.edu

PUBLICATIONS

SCI-Index Journal Publication

- [1] **Lin, Y. Z.**, Shi, Q., Yang, Z., Latibari, B. S., Shao, S., Salehi, S., & Satam, P. (2025). DDD-GenDT: Dynamic data-driven generative digital twin framework. *IEEE Transactions on Artificial Intelligence*.
- [2] Rahman, M. W. U., **Lin, Y. Z.**, Weeks, C., Ruddell, D., Gabriellini, J., Hayes, B., ... & Ziegler Jr, E. V. (2026). AI/ML-based detection and categorization of covert communication in IPv6 networks. *Cybersecurity*, 9(1), 33.
Note: Collaborated with the National Security Agency
- [3] Youssef, A., **Lin, Y.-Z.**, Satam, S., Saber Latibari, B., Pacheco, J., Salehi, S., Hariri, S., & Satam, P. (2025). Autonomous vehicle security: Hybrid threat modeling approach. *IEEE Open Journal of Vehicular Technology*.

- [4] Ghimire, S. *, **Lin, Y. -Z. ***, Mamun, M., Chowdhury, M. A., Alemi, F., Cai, S., ... & Salehi, S. (2025). HWREx: AI-enabled Hardware Weakness and Risk Exploration and Storytelling Framework with LLM-assisted Mitigation Suggestion. *ACM Transactions on Design Automation of Electronic Systems*.
Note: *Authors contributed equally to this research.
- [5] Chuang, J. Y., **Lin, Y. Z.**, Hsiao, Y. M., Liu, Y. C., Shu, D., Kuan, C. K., ... & Yen, J. Y. (2020). Discussion and improvement of a blade-type XBPM with coupling suppression by compensating calibration coefficients. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 953, 163174.
Note: Collaborated with the Berkeley National Laboratory, and Argonne National Laboratory

Peer-Review Book Chapter

- [1] **Lin, Y. Z.**, Patel, K., Alhamadah, A. H. J., Shih, B. P. J., Redondo, M. W., Corona, D. R. V., ... & Satam, P. (2025). PRISM: A Personalized, Rapid, and Immersive Skill Mastery Framework for Personalizing Experiential Learning Through Generative AI. In *Generative AI in Higher Education Assessment: Theory, Practice, and Ethical Implications* (pp. 45-69). Cham: Springer Nature Switzerland.

Peer-Review Conference Paper

- [1] **Lin, Y. Z.**, Shih, B. P. J., Encinas, J. P. M., Achom, E. V. A., Patel, K. H., Pacheco, J. H., ... & Satam, P. (2026). LLM-MC-Affect: LLM-Based Monte Carlo Modeling of Affective Trajectories and Latent Ambiguity for Interpersonal Dynamic Insight. *arXiv preprint arXiv:2601.03645*. **(Accepted by ACL 2026 Main Conference, Top 50% of accepted papers)**
- [2] **Lin, Y.-Z.**, Ghimire, S., Nandimandalam, A., Camacho, J. M., Tripathi, U., Macwan, R., Shao, S., Rafatirad, S., Yasaei, R., Satam, P., & Salehi, S. (2025). LLM-HyPZ: Hardware vulnerability discovery using an LLM-assisted hybrid platform for zero-shot knowledge extraction and refinement. Proceedings of the IEEE International Conference on Data Mining Workshops (ICDMW)
- [3] **Lin, Y. Z.**, Petal, K., Alhamadah, A. H., Ghimire, S., Redondo, M. W., Corona, D. R. V., ... & Satam, P. (2025). Personalized Education with Generative AI and Digital Twins: VR, RAG, and Zero-Shot Sentiment Analysis for Industry 4.0 Workforce Development. In *2025 ASEE Annual Conference & Exposition*.
- [4] **Lin, Y. Z.**, Shao, S., Rahman, M. H., Shafae, M., & Satam, P. (2023, October). DT4I4-Secure: Digital Twin Framework for Industry 4.0 Systems Security. In *2023 IEEE 14th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)* (pp. 0200-0209). IEEE.
- [5] **Lin, Y. Z.**, Chuang, J. Y., Sheng, I. C., Cheng, Y. T., Chang, C. C., Yang, Y. C., ... & Huang, C. H. (2018). Development of a task-oriented chatbot application for monitoring Taiwan photon source front-end system. *Proc. PCaPAC'18*, 228-229.

Pre-print Paper

- [1] **Lin, Y.-Z.**, Shih, B. P.-J., Chien, H.-Y. A., Satam, S., Pacheco, J. H., Shao, S., Salehi, S., & Satam, P. (2026). Exploring silicon-based societies: An early study of the Moltbook agent community. *arXiv preprint arXiv:2602.02613*

Peer-Review Conference Paper (Co-Author)

- [1] Raul, G., **Lin, Y. Z.**, Patel, K., Shih, B. P. J., Redondo, M. W., Latibari, B. S., ... & Satam, P. (2025). RAG-PRISM: A Personalized, Rapid, and Immersive Skill Mastery Framework with Adaptive Retrieval-Augmented Tutoring. *arXiv preprint arXiv:2509.00646*. **(Accepted by IEEE Frontiers in Education Conference 2025)**
- [2] Patel, K., **Lin, Y. Z.**, Raul, G., Shih, B. P. J., Redondo, M. W., Latibari, B. S., ... & Satam, P. (2025). Integrating Generative AI into Cybersecurity Education: A Study of OCR and Multimodal LLM-assisted Instruction. *arXiv preprint arXiv:2509.02998*. **(Accepted by IEEE Frontiers in Education Conference 2025)**
- [3] Alhamadah, A., Mamun, M., Harms, H., Redondo, M., **Lin, Y. Z.**, Pacheco, J., ... & Satam, P. (2024). Photogrammetry for Digital Twinning Industry 4.0 (I4) Systems. In *2024 IEEE/ACS 21st International Conference on Computer Systems and Applications (AICCSA)* (pp. 1-6). IEEE
- [4] Zhang, C., Shao, W., Wang, X., Cao, Y., Alhamadah, A. H. J., **Lin, Y. Z.**, ... & Watkins, L. (2024, September). Explainable Autonomic Cybersecurity System for Smart Power Grid. In *2024 IEEE Conference on Communications and Network Security (CNS)* (pp. 1-6). IEEE.
- [5] Yang, Y. C., **Lin, Y. Z.**, Chuang, J. Y., Chan, C. K., & Chang, C. C. (2020, August). Status of the TPS Vacuum Control System. In *17th International Conference on Accelerator and Large Experimental Physics Control Systems (ICALEPCS'19), New York, NY, USA, 05-11 October 2019* (pp. 1485-1487). JACOW Publishing, Geneva, Switzerland.
- [6] Chuang, J. Y., Hsiao, Y. M., **Lin, Y. Z.**, Liu, Y. C., Cheng, Y. T., Shueh, C., ... & Yang, Y. C. (2019, November). Real-Time Synchronized Calibration and Computing System with EPICS Based Distributed Controls in the TPS XBPM System. In *8th International Beam Instrumentation Conference (IBIC'19), Malmö, Sweden, 08-12 September 2019* (pp. 548-551). JACOW Publishing, Geneva, Switzerland.
- [7] Yang, Y. C., Chuang, J. Y., Hsiao, Y. M., **Lin, Y. Z.**, Chan, C. K., & Chang, C. C. (2019). Vacuum Control System for the Taiwan Photon Source. *Proc. PCaPAC'18*, 63-65.
- [8] Yang, Y. C., Liang, C. C., Chen, C. S., Chung, J. Y., Chan, C. K., Chang, C. C., ... & **Lin, Y. Z.** (2018, June). Vibration Measurements in the TPS Vacuum System. In *9th Int. Particle Accelerator Conf.(IPAC'18), Vancouver, BC, Canada, April 29-May 4, 2018* (pp. 3772-3774). JACOW Publishing, Geneva, Switzerland.
- [9] Huang, C. H., Chiu, P. C., Wu, C. Y., **Lin, Y. Z.**, Hsiao, Y. M., Chuang, J. Y., ... & Hsu, K. T. (2018, March). Beam stability diagnostics with X-ray beam position monitor in the Taiwan Photon Source. In *6th Int. Beam Instrumentation Conf. (IBIC'17), Grand Rapids, MI, USA, 20-24 August 2017* (pp. 217-220). JACOW, Geneva, Switzerland.

- [10] Chuang, J. Y., **Lin, Y. Z.**, Cheng, C. M., Yang, Y. C., Chang, C. C., & Sheng, I. C. (2018). The Development of a Fpga Based Front End Safety Inter-Lock System.
- [11] Chuang, J. Y., **Lin, Y. Z.**, Chen, W. C., & Lin, C. S. (2017, December). The construction of coherence microscope for extreme ultraviolet mask defect inspection in synchrotron facility. In *2017 IEEE/SICE International Symposium on System Integration (SII)* (pp. 440-443). IEEE.
- [12] Chuang, J. Y., Kuan, C. K., Sheng, I. C., **Lin, Y. Z.**, Hsiao, Y. M., Yang, Y. C., ... & Lin, C. S. (2017, May). Development and Construction of Safety and Control Systems for the TPS Front End Interlock. In *8th Int. Particle Accelerator Conf.(IPAC'17), Copenhagen, Denmark, 14-19 May, 2017* (pp. 1947-1950). JACOW, Geneva, Switzerland.